IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MATHEMATICS FOR ENGINEERS IIA

Tuesday, 9 January 10:00 am

Time allowed: 1:30 hours

There are EIGHT questions on this paper with varying marks.

Answer all questions.

NO CALCULATORS ALLOWED

Any special instructions for invigilators and information for candidates are on page 1.

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ELEC50011A - MATHEMATICS 2A

The residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

1. Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \sqrt{2}e^{-i\pi/4}z - 2 + i).$$

A region R in the $z-$plane is bounded by the $x-$ and $y-$axes, and the lines $x = -1$, and $y = 2$. Determine the region R^* of the $w-$plane into which R is mapped under the transformation given by $w(z)$, explaining your reasoning. Sketch both regions R and R^*. [4]

2. A contour integral is given as

$$I = \oint_C \frac{e^{iz}}{z(z^2+1)}\,dz\,.$$

(a) Find the residues for the poles of $F(z) = \dfrac{e^{iz}}{z(z^2+1)}$ at $z = 0, i$, and use Cauchy's Residue Theorem to obtain I when the contour C consists of four parts: H_R, a semi-circular arc of radius R, in the upper half-plane, centred at the origin, the real intervals $[-R,-r]$ and $[r,R]$, and H_r: a semi-circular indentation of radius r, centred at the origin. You should *choose* H_r to be an indentation into the upper or the lower half-plane, and include a sketch of C. [3]

(b) Write I as a sum of four integrals, along the four parts of C, use Jordan's Lemma to show the integral along H_R vanishes as $R \to \infty$, and obtain the integral along *your choice* of H_r, as $r \to 0$. [4]

(c) Finally, use all the above results to obtain [4]

$$\int_{-\infty}^{+\infty} \frac{\sin x}{x(x^2+1)}\,dx\,.$$

3. Given the integral

$$I = \int_0^{2\pi} \frac{1}{\sin\theta - i}\, d\theta$$

show that

$$I = \oint_C \frac{1}{f(z)}\, dz$$

where C is the unit circle in the complex plane, and $f(z)$ is a quadratic polynomial. Hence, or otherwise, obtain I. [5]

4. A set of data corresponds to measurements of a quantity x at times t, as follows:

t	x
0	7
1	3
3	4
5	2

Find the line of best fit $x = at + b$ for the above data, using the method of least squares, with $a, b \in \mathbb{R}$ to be determined. [7]

5. Prove that a set of n orthonormal vectors in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$. [4]

6. A matrix is given as

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

The columns of A form a basis for $\mathbb{R}^3$. Use Gram-Schmidt orthogonalization and the columns of A to find an orthonormal basis for $\mathbb{R}^3$. [6]

7. Given

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix},$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$. Show that the orthogonal diagonalization of the smaller of the two matrices AA^T or A^TA gives the singular values and one of the orthogonal matrices U, V. Then find the SVD, stating at each point the basis for a fundamental subspace of A. [10]

8. A digital image consisting of 2000×1000 pixels is compressed for transmission using the Singular Value Decomposition. The first r principal components are used, and the data transmitted is 15% of the original amount, to the nearest 1%. How many components were used? Explain your reasoning. [3]